

$$1)_{116} = (1110100)_2$$

$$1110100 \gg \gg 3$$

$$1110100 \text{ mishavad } 1110 = 2+4+8=14$$

$$2)(2431)_7 = 1+3*7+4*49+2*243=$$

$$0000000000000001 + 0000000000000011 * 0000000000000111 + 0000000000000110 * 0000000000011001 + 0000000000000010 * 0000000011110011$$

$$= 0000001001011100$$

$$-(2431)_7 = \text{two's complement } 111110110100011 + 1 = 111110110100100$$

$$(2044)_6 = 100 + 100*110 + 0*100100 + 10*11011000 = 111001100$$

$$0000000111001100$$

$$+ 111110110100100$$

$$= 11111101110000$$

$$\text{Two's complement} = 0000000010010000 = -144$$

$$3) (A2)_{13} * (12)_9 = (10 + 1010 * 1101) * (10 + 1 * 1001) = 0000010101001000 = 1452$$

```
2)float x, y = 0;  
x = 3 + 12 * 3 / 5 - 9 % 3;  
y = x % 5;(x=10.2 x%5 is meaningless)  
printf ("%d", x % y);
```

my “correction”:

```
#include<stdio.h>  
Void main()  
{  
int x;  
x = 3 + 12 * 3 / 5 - 9 % 3;  
float y;  
y = x % 5  
printf ("%d", x % y);  
}
```

```
include <stdio.h>
define Pi 3.1415;(should be placed below int main() )
int a = 5;
int main() (it should be placed at the beginning)
{
int a; b; c;
(the amount of b and c isn't given)
float result = (a - Pi + c) * b;
printf("%d", result);
}
```

My “correction”:

```
include <stdio.h>
void main()
{
int a; b; c;
a = 5;
b=3;
c=2;
define Pi 3;
float result = (a - Pi + c) * b;
printf("%d", result);
}
```

سوال 3 قسمت دوم

$124.133 \quad 321.432 \geq 123.133$

++ عدد را با یک جمع می کند. 3 بعد از % یعنی تا سه رقم اعشار عدد را نمایش
دهد. 4 یعنی 4 بایت حافظه اختصاص بده یعنی 32 بیت که کافیست.

سوال 4(محیط و مساحت)

```
#include<stdio.h>
```

```
#include<math.h>
```

```
void main()
```

```
int main()
```

```
{
```

```
float a, b, c; // inputs<
```

```
float perimeter , area; // outputs
```

```
/* mathematical statements */
```

```
float perimeter;
```

```
perimeter=a+b+c;
```

```
float d;
```

```
d=(a+b+c)/2-a;
```

```
float e;
```

```
e=(a+b+c)/2-b;
```

```
float f;
```

```
f=(a+b+c)/2-c;
```

```
float g;
```

```
g=perimeter*d*e*f;
```

```
float area;
```

```
g=area*area;
```

```
//shoma ra be khoda azlaee mosallas ra radicali nadahid;
```

```
printf("Area: %f \nPerimeter: %f", area , perimeter);
```

```
}
```